

Embryonic origins of forebrain oligodendrocytes revisited by combinatorial genetic fate mapping

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Abstract

Multiple embryonic origins give rise to forebrain oligodendrocytes (OLs), yet controversies and uncertainty exist regarding their differential contributions. We established intersectional and subtractional strategies to genetically fate map OLs produced by medial ganglionic eminence/preoptic area (MGE/POA), lateral/caudal ganglionic eminences (LGE/CGE) and dorsal pallium. We found that, contrary to the canonical view, LGE/CGE-derived OLs make minimum contributions to the neocortex and corpus callosum, but dominate piriform cortex and anterior commissure. Additionally, MGE/POA-derived OLs, instead of being entirely eliminated, make small but sustained contribution to cortex with a distribution pattern distinctive from those derived from the dorsal origin. Our study provides a revised and more comprehensive view of cortical and white matter OL origins, and established valuable new tools and strategies for future OL studies.

eLife assessment

In this study, the authors provide **valuable** evidence that the LGE is not a significant source of oligodendrocytes for the cortex. The reviewers did find some technical considerations that call for some modulation of the strength of the authors' conclusions and also pointed out some aspects of the data that were **incomplete** as presented.

Introduction

Oligodendrocytes (OLs) are an important class of macroglia responsible for producing the myelin sheaths that insulate and protect neuronal axons. Forebrain OLs arise from multiple embryonic origins. Previous fate mapping study using *Nkx2.1^{Cre}*^[1], *Gsh2^{Cre}*^[2] and *Emx1^{Cre}*^[3] reported consecutive and competing waves of OLs derived from medial ganglionic eminence/preoptic area (MGE/POA), lateral/caudal ganglionic eminences (LGE/CGE) and dorsal pallium^[2]. The first-wave of OLs generated by MGE/POA (_{MP}OLs) were believed to be eliminated postnatally, while those from the second and third waves (_{LC}OLs and _{dOL}s) survive and

populate the cortex and corpus callosum at comparable proportions. Several other studies provided both supporting and contradicting evidence to this model[4–9]. Moreover, *Gsh2* was recently found to be expressed in dorsal progenitors[10], casting doubt on the interpretation of lineage tracing data from *Gsh2^{Cre}*.

In this study, we generated new genetic tools and combinatorial fate mapping strategies which allow direct visualization and comparison among OLs derived from different origins. We found that neocortical OLs are primarily composed of dOLs, rather than similar proportions of _{LC}OLs and dOLs. In contrast, _{LC}OLs and dOLs made comparable contributions to piriform cortex. We also found that although _{MP}OLs only make a small contribution, they do persist in the cortex beyond adulthood with a unique spatial pattern distinct from that of the dOLs. In the two major white matter commissure tracts, dOLs are the vast majority in corpus callosum but make little contribution to anterior commissure, while _{LC}OLs behaved the opposite. These findings significantly revised the classical view and provided a new and more comprehensive picture of cortical and white matter OL origins.

Results and Discussion

To unambiguously track OLs from different embryonic origins, we first generated a knock-in driver, *Opalin^{P2A-Flpo-T2A-tTA2}* (Figure 1), orthogonal to Cre drivers that label dorsal or ventral progenitors (*Progenitor^{Cre}*). *Opalin* (also known as *Tmem10*) encodes oligodendrocytic myelin paranodal and inner loop protein that are specifically expressed in differentiated oligodendrocytes[11–14]. In *Opalin^{P2A-Flpo-T2A-tTA2}*, *Flpo* and *tTA2* were inserted before the STOP codon and linked by self-cleavage peptide P2A and T2A (Figure 1A–C), allowing co-transcription and translation with *Opalin*. Flp-mediated recombination by this driver (hereinafter referred to as *Opalin^{Flp}* for simplicity) enables highly specific, efficient and irreversible OL labeling, while the tTA2 component offers the flexibility for labeling in tunable densities (Figure 1D–F).

Next, we established two types of genetic combinatorial fate mapping strategies to directly visualize OLs from different embryonic origins (Figure 2): (1) combining *Opalin^{Flp}* and *Progenitor^{Cre}* with intersectional reporters *Ai65* to label OLs derived from Cre+ progenitor domain by RFP (Figure 2A); (2) combining *Opalin^{Flp}* and *Progenitor^{Cre}* with *RC::FLTG*[15] to simultaneously label OLs derived from Cre+ progenitors by GFP and OLs derived from the complementing Cre-progenitors by RFP (Figure 2B). The first approach allowed us to track dOLs and _{MP}OLs (Figure 2C, E). The second approach empowered us to observe and compare OLs generated from dorsal and ventral origins, or those from *Gsh2*+ and *Gsh2*-progenitors, in the same brain (Supplementary Figure 1). Importantly, the subtraction power enabled us to target OLs derived from LGE/CGE progenitors that express neither *Emx1* nor *Nkx2.1* (Figure 2D). In addition, these strategies greatly facilitated the identification of OLs derived from specific origin which exist at relatively low density in certain regions.

Deploying these strategies, we assessed the differential contributions of dOLs, _{LC}OLs and _{MP}OLs by analyzing RFP+ cells in the following mice: *Opalin^{Flp}::Emx1^{Cre}::Ai65* (Figure 2C), *Opalin^{Flp}::Emx1^{Cre}::Nkx2.1^{Cre}::RC::FLTG* (Figure 2D) and *Opalin^{Flp}::Nkx2.1^{Cre}::Ai65* (Figure 2E). We observed two significant differences from the traditional model in the neocortex. The first major deviation is that, instead of comparable contributions by dOLs and _{LC}OLs, the vast majority of neocortical OLs were dOLs but not _{LC}OLs. The density of dOLs is significantly higher than that of _{LC}COLs in the motor cortex (Mo) (504.5±25.1 vs. 5.3±0.7; P<0.001; n=3) and somatosensory cortex (SS) (518.0±22.6 vs. 21.6±2.3; P<0.001; n=3) (Figure 2C, D, F). Co-staining with aspartoacylase (ASP) [16], a marker for mature OL, also revealed a very high neocortical OL labeling rate in *Opalin^{Flp}::Emx1^{Cre}::Ai65* (RFP+ASP+/ASP+: 94.6±2.0% in Mo and 90.5±2.2% in SS; n=3). Considering the possibility of incomplete recombination in combinatorial reporters, and

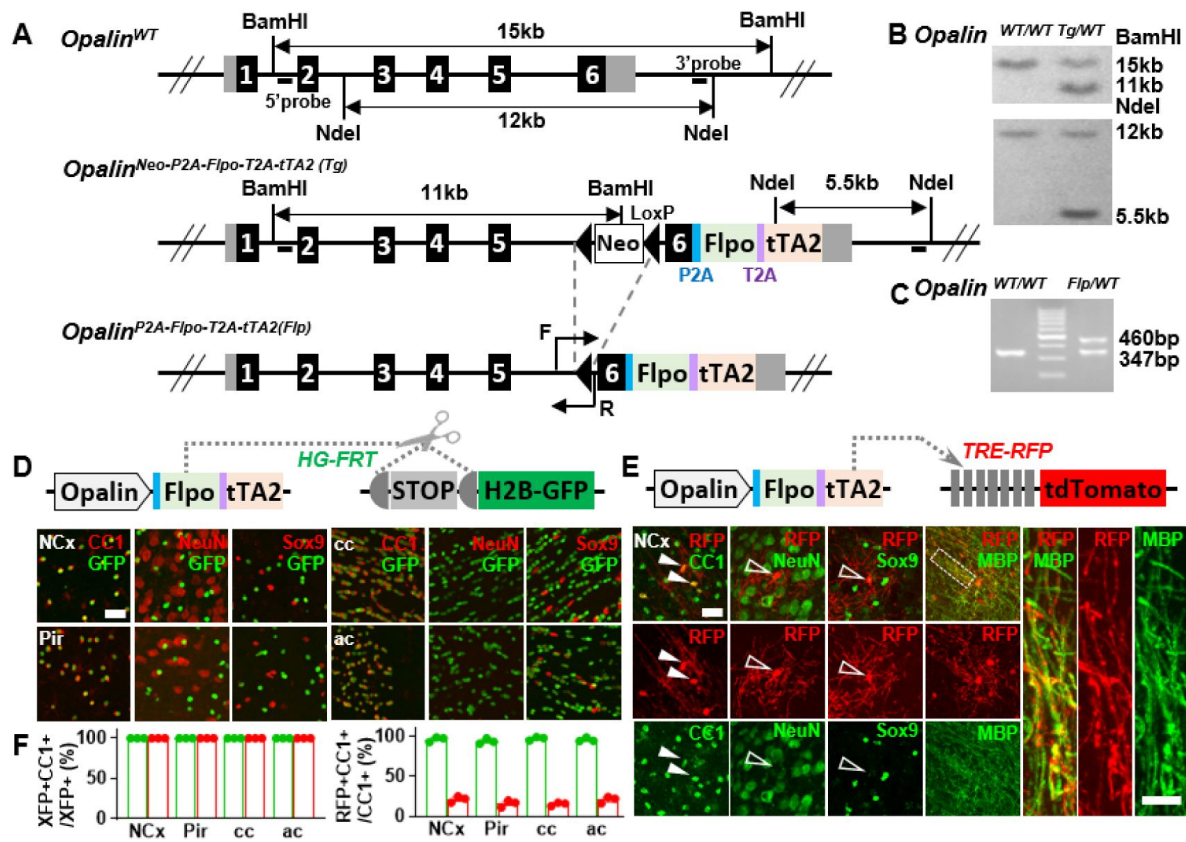


Figure 1.

A new driver mouse for efficient and specific OL labeling

(A) Scheme for generating the *Opalin*^{P2A-Flpo-T2A-tTA2} allele. (B) Southern blot confirmation of correctly targeted ES clone. (C) Genomic PCR to genotype F1 offspring. (D) OL labeling by Flp. (E) OL labeling by tTA2. High magnification images of the boxed region showing co-localization of RFP with MBP staining, which further demonstrated the myelination ability of labeled OLs. (F) Quantification of marker staining. Left panel quantified the labeling specificity. Both systems are highly specific, as shown by the complete co-localization with OL marker (CC1) and lack of co-staining by the neuronal marker (NeuN) or astrocyte marker (Sox9). Right panel quantified the labeling efficiency. Close to complete OL labeling was achieved by Flp-dependent H2B-GFP reporter in all analyzed regions, while sparser labeling with variable regional density was achieved by tTA2-dependent tdTomato reporter driven by TRE promoter. Scale bar: 50 μ m in low magnification images, 5 μ m in high magnification images. Quantification: n=3. Dots represent data from individual mice.

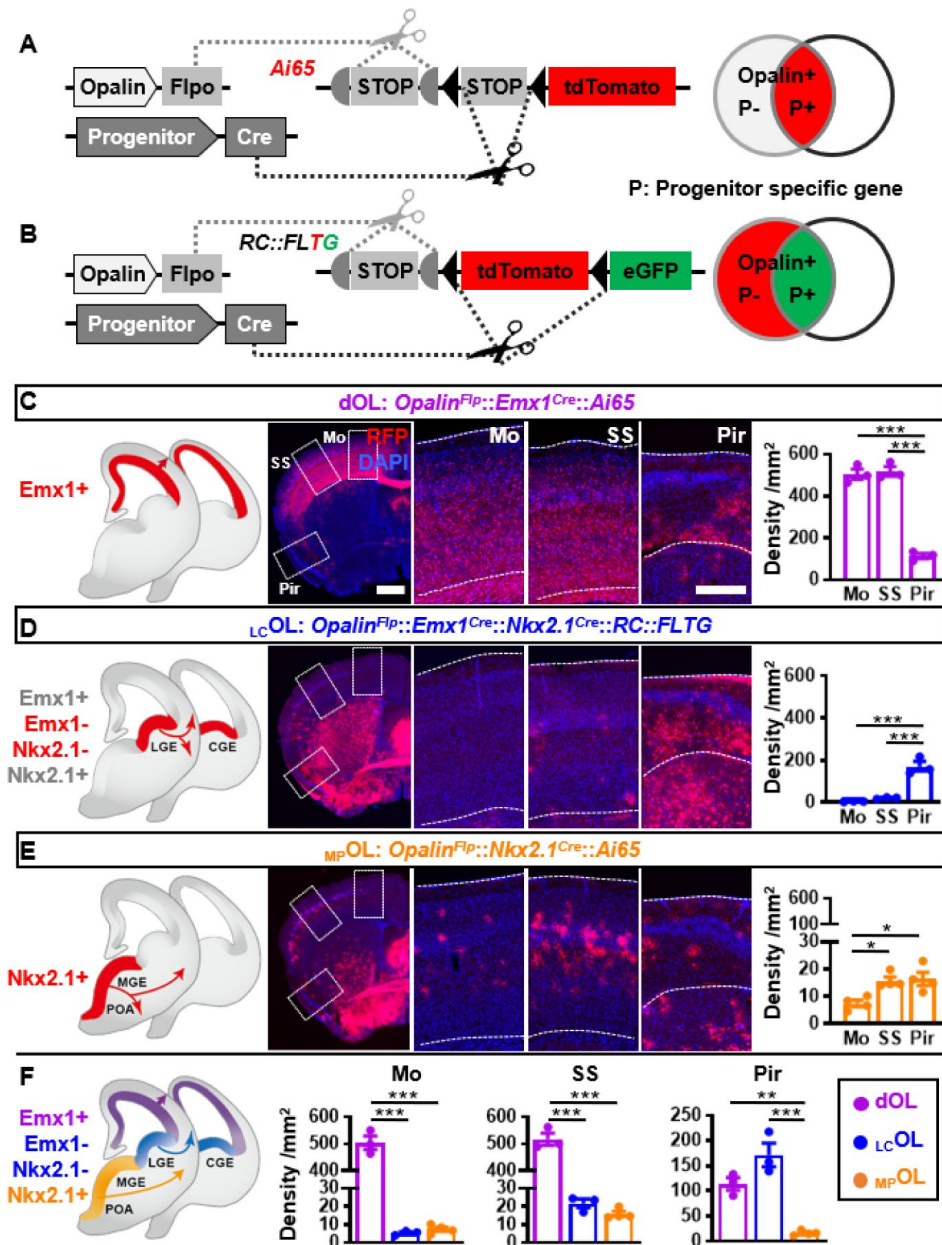


Figure 2.

Combinatorial fate mapping of dOLs, MPOLs and LCOLs

(A) Strategy for intersectional labeling. Flp-AND-Cre labels OLs from Cre-expressing progenitors with RFP. (B) Strategy for simultaneous intersectional and subtractive labeling of OLs derived from complementary origins. Flp-NOT-Cre labels OLs from non-Cre-expressing progenitors with RFP, while Flp-AND-Cre labels OLs from Cre-expressing progenitors with GFP. (C-E) Intersectional labeling of dOLs in *Opalin^{Flpo}::Emx1^{Cre}::Ai65*, subtractive labeling of LCOLs in *Opalin^{Flpo}::Emx1^{Cre}::Nkx2.1^{Cre}::RC::FLTG*, and intersectional labeling of MPOLs in *Opalin^{Flpo}::Nkx2.1^{Cre}::Ai65* and their cortical distribution. Schematics, representative images and quantification of RFP+ cell density in motor cortex (Mo), somatosensory cortex (SS) and piriform cortex (Pir) were displayed. (F) Differential contribution of OLs from the three embryonic origins to Mo, SS and Pir.

Scale bar: 1mm in low magnification images, 500 μ m in high magnification images of the boxed area. Quantification: n=3 for dOLs and LCOLs; n=4 for MPOLs. Dots represent data from individual mice. **P < 0.01, ***P < 0.001. Error bar: S.E.M.

the relatively low Cre activity in the dorsal MGE of *Nkx2.1^{Cre}* [1], the genuine contribution of L_C OLs to the neocortex could be even lesser than our current observation. Therefore, the large quantity of neocortical OLs labeled by *Gsh2^{Cre}* in previous study [2] or by GFP in *Opalin^{Flp}::Gsh2^{Cre}::RC::FLTG* (Supplementary Figure 1B) most likely were predominantly dOLs generated by Gsh2+ dorsal progenitors [10], rather than bona fide L_C OLs.

The second major deviation is that cortical M_P OLs are not completely depleted postnatally, instead, they make a small but continued contribution with a unique spatial distribution pattern (Figure 2D and Supplementary Figure 2). M_P OLs display a clear rostrocaudal density decline (Supplementary Figure 2A-B) and a laminar preference towards layer 4 (L4) in SS (Supplementary Figure 2C-E). In contrast, the distribution of dOLs does not vary significantly across the rostrocaudal axis but exhibits increased density towards deeper layers (Supplementary Figure 2A-E). Importantly, we have observed cortical M_P OLs in mice as old as one year (Supplementary Figure 2F), well beyond the age analyzed in previous reports [2, 7, 8], suggesting a persisted contribution.

We then turned our attention to the lateral three-layer archicortex, piriform cortex (Pir). Different from the neocortex, Pir contains comparable density of dOLs and L_C OLs (114.1 ± 12.5 vs. 171.7 ± 23.6 ; $P=0.058$; $n=3$) (Figure 2F). M_P OLs make the lowest contribution at a density similar to SS (16.4 ± 2.4 vs. 15.8 ± 1.4 ; $P=0.966$; $n=4$) (Figure 2E).

These combinatorial models also grant us the opportunity to revisit the differential contributions of dOLs, L_C OLs and M_P OLs to the two commissural white matter tracts, corpus callosum (cc) and anterior commissure (ac), which contain higher density of OLs than the grey matter (Figure 3). We found that, similar to the neocortex, cc is mainly populated by dOLs and supplemented by very low density of L_C OLs and M_P OLs (Figure 3A-D). Interestingly, L_C OLs and M_P OLs seem to show preferential distribution in the lateral and medial regions of cc (cc-l and cc-m), respectively (Figure 3B-D, boxed regions in right panels). Different from cc, ac is mainly populated by L_C OLs (2454.6 ± 254.1 , $n=3$) and M_P OLs (1146.3 ± 101.6 ; $n=4$) and supplemented by very low density of dOLs (85.2 ± 34.5 ; $n=3$) (Figure 3D).

To substantiate the above results, we further breed *Opalin^{Flp}::Emx1^{Cre}::Nkx2.1^{Cre}::Ai65* to label dOLs together with M_P OLs by RFP and co-stained them with ASPA (Supplementary Figure 3). RFP-ASPA+ cells were difficult to find in Mo, SS and cc, but were more easily observed in Pir and ac, consistent with the respective low and high L_C OL densities in these regions.

In summary, our findings significantly revised the canonical model of forebrain OL origins (Figure 4A), and provided a new and more comprehensive view (Figure 4B). We demonstrated that neocortical OLs are mainly derived from dorsal origin with small but lasting contribution from the ventral origin (Figure 2, Supplementary Figure 1A). Our data showed that LGE/CGE makes little contribution to neocortex and cc, but makes major contribution to piriform cortex and ac. This finding is supported by another report [17] in which in utero electroporation failed to label LGE-derived cortical OLs. Taken together, these results suggest that the lack of L_C OL in neocortex is less likely caused by competitive postnatal elimination, but more likely due to limited production and/or allocation. We further discovered that MGE/POA makes a small but persistent contribution to the neocortex with a distinct distribution pattern featured by a rostral-high to caudal-low gradient and a preference towards L4 in SS. Whether their enduring existence and highly biased localization has functional implications awaits future exploration. In addition, we found that the cc showed a similar OL composition as the neocortex, but the Pir and the ac each exhibited distinct OL compositions in term of their embryonic origins. L_C OLs are the major contributor to both regions, while dOLs and M_P OLs mainly contribute to Pir and ac, respectively.

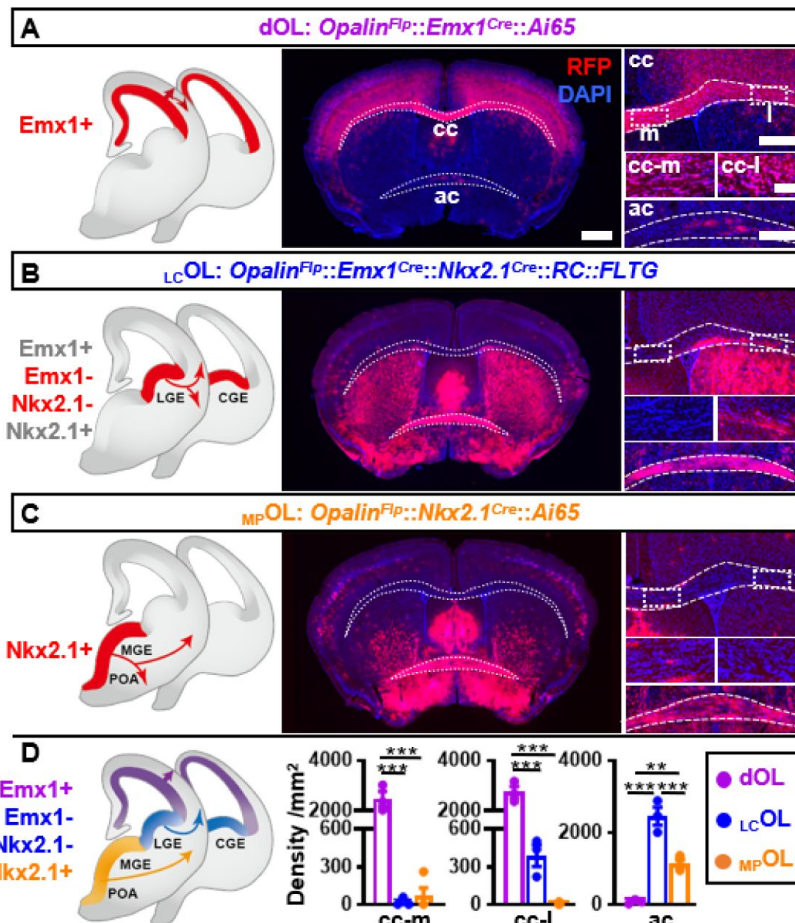


Figure 3.

dOLs, MPOLs and LCOLs differentially contribute to the two major white matter tracts

(A-C) Schematics and representative images of dOLs, MPOLs and LCOLs in the two major commissure white matter tracts: corpus callosum(cc) and anterior commissure (ac). MPOLs and LCOLs preferentially reside in the medial and lateral cc (cc-m and cc-l), respectively. (D) Quantification of dOL, MPOL and LCOL densities in cc-m, cc-l and ac. Scale bar: 1mm in low magnification images, 500µm in cc and ac, 100 µm in high magnification images of the boxed area (cc-m and cc-l) in cc. Quantification: n=3 for dOLs and LCOLs; n=4 for MPOLs. Dots represent data from individual mice. **P < 0.01, ***P < 0.001. Error bar: S.E.M.

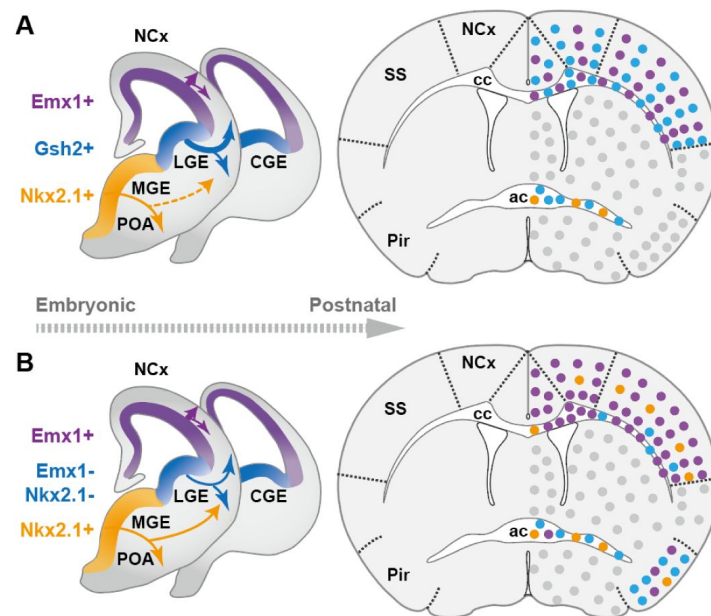


Figure 4.

The classical and revised model of forebrain OL origins

(A) In the classical model [2], OLs derived from MGE/POA (orange) were largely eliminated postnatally (thin dashed line), while those from LGE/CGE (blue) and dorsal origin (purple) survive at similar proportions (thick solid line). Therefore, neocortex (NCx) and corpus callosum (cc) contain comparable density of LcOLs (blue dots) and dOLs (purple dots) and are devoid of MpOLs (orange dots). (B) In the new model, NCx and cc mainly contains dOLs with very low contribution from the ventral origins. LcOLs mainly contribute to Pir and ac. MpOLs make a small but sustained contribution to NCx, with a strong laminar preference towards layer 4 in somatosensory cortex (SS). In addition, dOLs and MpOLs also make substantial contributions to Pir and ac, respectively. Grey dots indicate OLs in unanalyzed regions.

In addition to the new framework of forebrain OL origins (**Figure 4**), we also generated a new driver (**Figure 1**) and established multiple combinatorial genetic models (**Figure 2**) for efficient tracking and direct visualization of OLs from different embryonic origins without interference from other cell types. These tools set up a firm foundation and will provide reliable experimental access for future inquiries on the development and function of diverse OLs in healthy and disease brains[18], especially to uncover the relationship between developmental origins and the functional and molecular heterogeneity of OLs.

Material and Methods

Mice

All mouse studies were carried out in strict accordance with the guidelines of the Institutional Animal Care and Use Committee of Fudan University. All husbandry and experimental procedures were reviewed and approved by the same committee. All applicable institutional and/or national guidelines for the care and use of animals were followed. The following transgenic mouse lines were used in this study: *Nkx2.1^{Cre}* (Jax 008661)[1], *Gsh2^{Cre}* (Jax 025806)[2], *Emx1^{Cre}* (Jax 005628)[3], *Ai65* (Jax 021875)[19], *RC::FLTG* (Jax 026932)[15]. The tTA2-dependent tdTomato reporter (*TRE-RFP*) was derived from *Ai62* (Jax 022731) [19], by removing LoxP-STOP-LoxP with *E2a-Cre* (Jax 003724). The Flp dependent H2B-GFP reporter (*HG-FRT*) was derived from *HG-dual* (Jax 028581) via removal of loxP flanking STOP cassette by *CMV-Cre*[20]. The *Opalin^{P2A-Flpo-T2A-tTA2}* allele was generated by targeted insertion of the T2A-Flpo-P2A-tTA2 sequence immediately before the STOP codon of the endogenous *Opalin* gene using homologous recombination. Gene targeting vector was generated using PCR-based cloning approach as described before[20]. More specifically, a 4.7 kb 5' homology arm, a loxP flanking Neo positive selection cassette, a T2A-Flpo-P2A-tTA2 cassette and a 2.7 kb 3' homology arm were cloned into a building vector containing the DTA negative selection cassette to generate the targeting vector. Targeting vector was linearized and transfected into a C57/black6 ES cell line. ES clones that survived through negative and positive selections were first screened by genomic PCR, then confirmed by Southern blotting using appropriate DIG-dUTP labelled probes. One positive ES cell clone was used for blastocyst injection to obtain male chimera mice carrying the modified allele following standard procedures. Chimera males were bred with C57BL/6J females to confirm germline transmission by genomic PCR. The Neo selection cassette was self-excised during spermatogenesis of F0 chimeras. Heterozygous F1 siblings were bred with one another to establish the colony. Targeting vector construction, ES cell transfections and screening, blastocyst injections and chimera breeding were performed by Biocytogen Co, China.

Genomic PCR

Genomic DNA was prepared from mouse tails. Tissue was lysed by incubation in tail lysis buffer (Viagen, 102-T) with 0.1 mg/ml proteinase K (Diamond, A100706) overnight at 55°C followed by 45 min at 90°C in an air bath to inactivate proteinase K. The lysate was cleared by centrifugation at maximum speed (21,130 G) for 15 min in a table-top centrifuge. Supernatant containing genomic DNA was used as the PCR template for amplifying DNA products. The following primers were used:

Opalin-F: 5'-GGCCTATGTTTGATTTCAGCACTG-3'

Opalin-R: 5'-AGCACTTATGACTGCTGAGCCGTTTC-3'

Immunohistochemistry and microscopy

Mice were anaesthetized by intraperitoneal injection of 1.5% sodium pentobarbital (0.09 mg/g body weight) and then intracardially perfused with saline followed by 4% paraformaldehyde in 0.1 M PB. Following post fixation at 4°C for 24 hours, brain samples were sectioned at 30 µm using a vibratome (Leica VT1000S), or transferred into 30% sucrose in 0.1M PB for cryoprotection, OCT embedded, and sectioned using a cryostat (Leica CM1950). For CC1 immunostaining, antigen retrieval was performed prior to blocking by boiling for 3 min in 10 mM citrate buffer (pH6.0). Sections were blocked in PBS containing 0.05% Triton and 5% normal donkey serum and then incubated with the following primary antibodies in the blocking solution at 4 °C overnight: RFP (goat polyclonal antibody, 1:2000, SICGEN AB0081-200; rabbit polyclonal antibody, 1:2000, Rockland 600-401-379), GFP (chicken polyclonal antibody, 1:1000, Aves Labs, GFP-1020), MBP (rat polyclonal antibody, 1:500, AbD Serotec, MCA409S), CC1 (rabbit polyclonal antibody, 1:500, Oasis Biofarm, OB-PRB070, mouse polyclonal antibody, 1:300, Millipore, OP80), ASPA (rat polyclonal antibody, 1:200, Oasis Biofarm, OB-PRT005), Sox9 (rabbit polyclonal antibody, 1:2000, Chemicon, AB5535), NeuN (mouse monoclonal antibody, 1:500, Millipore, MAB377). Sections were then incubated with appropriate Alexa fluor dye-conjugated IgG secondary antibodies (1:500, ThermoFisher Scientific) or CF dye-conjugated IgG secondary antibodies (1:250, Sigma) in blocking solution and mounted in Aqua-mount (Southern Biotech, 0100-01). Sections were counterstained with DAPI. Sections were imaged with confocal microscopy (Olympus FV3000), fluorescence microscopy (Nikon Eclipse Ni; Olympus VS120; Olympus VS200) and fluorescent stereoscope (Nikon SMZ25). Anatomical regions in the postnatal brain were identified according to the *Paxinos "The Mouse Brain" Atlas* and the *Allen Reference Atlas*, and their areas were measured in ImageJ, whenever applicable. One out of every four sections within the range of selection from each brain and at least three brains per genotype were analyzed. To quantify density and co-localization, cells were identified and counted in Image J or QuPath.

Statistical analysis

GraphPad Prism version 8.0.1 was used for statistical calculations. No statistical methods were used to predetermine sample sizes, but our sample sizes are similar to those reported in previous publications. Data collection and analysis were performed blind to the conditions of the experiments whenever possible. No animals or data points were excluded from the analysis. Normalcy was assessed using Shapiro-Wilk test. Equal variances were assessed using F test or Bartlett's test. Statistical significance was tested using two-tailed unpaired t-test, Welch's t-test, one-way ANOVA and two-way ANOVA followed by Tukey's or Bonferroni post hoc test, wherever appropriate. Data are presented as mean ± standard error of the mean (S.E.M.). $P < 0.05$ was considered significant. Significance is marked as * $P < 0.05$; ** $P < 0.01$ and *** $P < 0.001$.

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Investigation: YC, ZZ, MS

Visualization: YC, MH, LG

Supervision: MH, LG

Writing—original draft: MH, YC, LG

Writing—Review and editing: MH, YC, ZZ, MZ

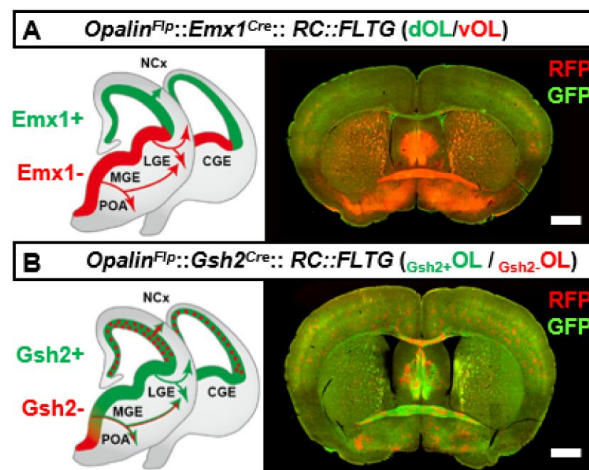
Declaration of interests

The authors declare no competing interests.

Supplementary information

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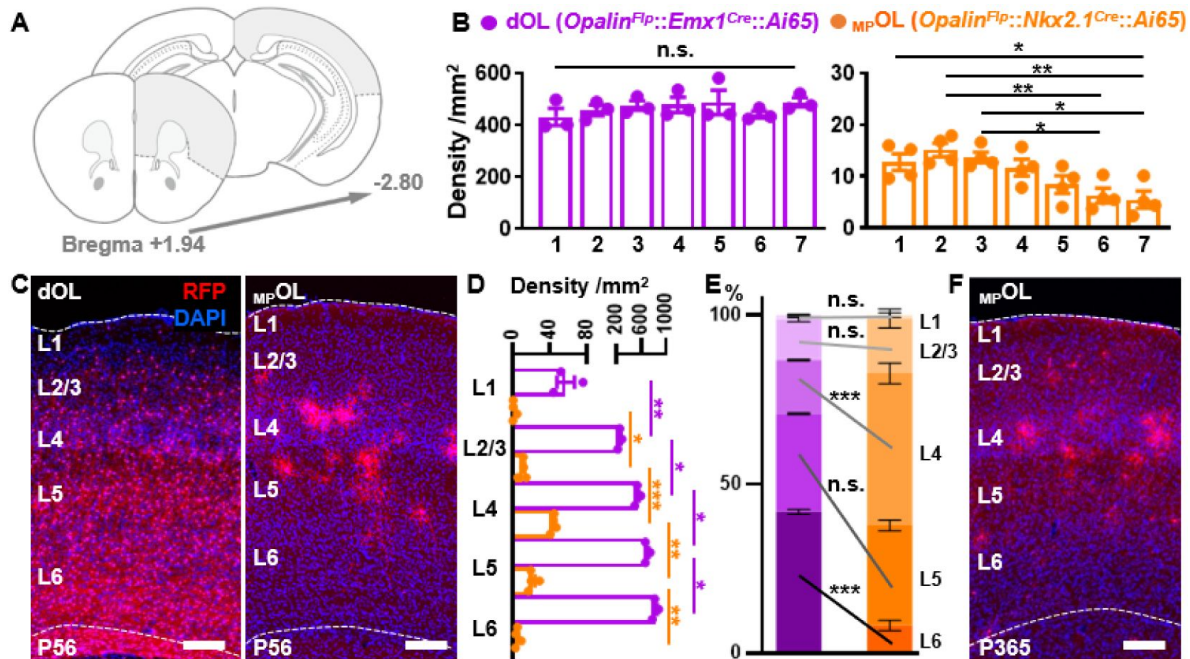
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Supplementary Figure 1.

Simultaneous differential labeling of OLs derived from complementary embryonic origins

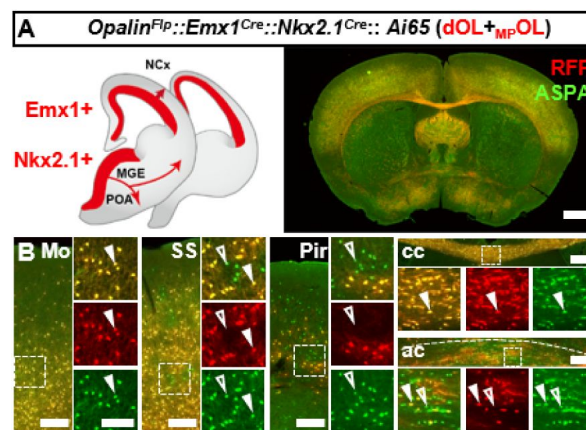
(A) Coronal sections showing GFP+ OLs from dorsal origin (dOLs) and RFP+ OLs from ventral origin (vOLs) in *Opalin^{Flp}::Emx1^{Cre}::RC::FLTG*. (B) Coronal sections showing GFP+ OLs derived from Gsh2+ progenitors (G_{sh2+} OL) and RFP+ OLs derived from Gsh2-progenitors (G_{sh2-} OL) in *Opalin^{Flp}::Gsh2^{Cre}::RC::FLTG*. Scale bar: 1mm.



Supplementary Figure 2.

The distribution pattern of neocortical dOLs and MPOLs

(A-B) Rostrocaudal distribution of neocortical dOLs and MPOLs. (A) Within the range of analysis, RFP+ OLs in neocortex of every fourth section were quantified. (B) Slices were grouped into seven bins numbered rostrocaudally. Density of dOLs showed no significant change across bins, while MPOLs exhibited lower density in more caudal regions. (C-E) Laminar distribution of dOLs and MPOLs in somatosensory cortex (SS). (C) Representative images of SS from P56 mice. (D) Distribution of dOLs and MPOLs across 6 layers. More dOLs reside in deeper layers, while MPOLs are highly enriched in L4. (E) dOLs and MPOLs exhibited significant differences in their relative laminar distributions. (F) Representative image of SS from 1 year old *Opalin^{Flp}::Nkx2.1^{Cre}::Ai65* mouse. Scale bar: 200 μ m. Dots represent data from individual mice. Error bar: S.E.M. *P < 0.05, **P < 0.01, ***P < 0.001.



Supplementary Figure 3.

Intersectional labeling of OLs derived from both dorsal origin and MGE/POA

(A) Coronal sections showing both dOLs and _{MP}OLs labeled by RFP in *Opalin^{Flp}::Emx1^{Cre}::Nkx2.1^{Cre}::Ai65*. (B) Higher magnification images showing ASPA+ RFP+ dOLs/_{MP}OLs (closed arrow heads) and ASPA+RFP-putative _LCOLs (open arrow heads). The latter cells were difficult to find in neocortical regions such as motor cortex (Mo) and somatosensory cortex (SS), and corpus callosum (cc), but were frequently encountered in piriform cortex (Pir) and anterior commissure (ac). Scale bar: 1mm in low magnification images, 200µm in Mo, SS, cc and ac, 10 µm in high magnification images of boxed area (all shared the same scale).

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Reviewer #1 (Public Review):

Summary:

In this study, the authors generated a novel transgenic mouse line OpalinP2A-Flpo-T2A- τ TA2 to specifically label mature oligodendrocytes, and at the same time their embryonic origins by crossing with a progenitor cre mouse line. With this clever approach, they found that LGE/CGE-derived OLs make minimum contributions to the neocortex, whereas MGE/POA-derived OLs make a small but lasting contribution to the cortex. These findings are contradictory to the current belief that LGE/CGE-derived OPCs make a sustained contribution to cortical OLs, whereas MGE/POA-derived OPCs are completely eliminated. Thus, this study provides a revised and more comprehensive view on the embryonic origins of cortical oligodendrocytes. To specifically label mature oligodendrocytes, and at the same time their embryonic origins by crossing with a progenitor cre mouse line. With this clever approach, they found that LGE/CGE-derived OLs make minimum contributions to the neocortex, whereas MGE/POA-derived OLs make a small-but-lasting contribution to to cortex. These findings are contradictory to the current belief that LGE/CGE-derived OPCs make a sustained

contribution to cortical OLs, whereas MGE/POA-derived OPCs are completely eliminated. Thus, this study has provided a revised and updated view on the embryonic origins of cortical oligodendrocytes.

Strengths:

The authors have generated a novel transgenic mouse line to specifically label mature differentiated oligodendrocytes, which is very useful for tracing the final destiny of mature myelinating oligodendrocytes. Also, the authors carefully compared the distribution of three progenitor cre mouse lines and suggested that Gsh-cre also labeled dorsal OLs, contrary to the previous suggestion that it only marks LGE-derived OPCs. In addition, the author also analyzed the relative contributions of OLs derived from three distinct progenitor domains in other forebrain regions (e.g. Pir, ac). Finally, the new transgenic mouse lines and established multiple combinatorial genetic models will facilitate future investigations of the developmental origins of distinct OL populations and their functional and molecular heterogeneity.

Weaknesses:

Since OpalinP2A-Flpo-T2A-tTA2 only labels mature oligodendrocytes but not OPCs, the authors can not suggest that the lack of LGE/CGE-derived-OLs in the neocortex is less likely caused by competitive postnatal elimination, but more likely due to limited production and/or allocation (line 118-9). It remains possible that LGE/CGE-derived OPCs migrate into the cortex but are later eliminated.

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Reviewer #2 (Public Review):

Summary:

In this manuscript, Cai et al use a combination of mouse transgenic lines to re-examine the question of the embryonic origin of telencephalic oligodendrocytes (OLs). Their tools include a novel Flp mouse for labelling mature oligodendrocytes and a number of pre-existing lines (some previously generated by the last author in Josh Huang's lab) that allowed combinatorial or subtractive labelling of oligodendrocytes with different origins. The conclusion is that cortically-derived OLs are the predominant OL population in the motor and somatosensory cortex and underlying corpus callosum, while the LGE/CGE generates OLs for the piriform cortex and anterior commissure rather than the cerebral cortex. Small numbers of MGE-derived OLs persist long-term in the motor, somatosensory and piriform cortex.

Strengths:

The strength and novelty of the manuscript lies in the elegant tools generated and used and which have the potential to elegantly and accurately resolve the issue of the contribution of different progenitor zones to telencephalic regions.

Weaknesses:

(1) Throughout the manuscript (with one exception, lines 76-78), the authors quantified OL densities instead of contributions to the total OL population (as a % of ASPA for example). This means that the reader is left with only a rough estimation of the different contributions.

(2) All images and quantifications have been confined to one level of the cortex and the potential of the MGE and the LGE/CGE to produce oligodendrocytes for more anterior and more posterior cortical regions remains unexplored.

(3) Hence, the statement that "In summary, our findings significantly revised the canonical model of forebrain OL origins (Figure 4A) and provided a new and more comprehensive view (Figure 4B)." (lines 111, 112) is not really accurate as the findings are neither new nor comprehensive. Published manuscripts have already shown that (a) cortical OLs are mostly generated from the cortex [Tripathi et al 2011 (<https://doi.org/10.1523/JNEUROSCI.6474-10.2011>), Winker et al 2018 (<https://doi.org/10.1523/JNEUROSCI.3392-17.2018>) and Li et al (<https://doi.org/10.1101/2023.12.01.569674>)] and (b) MGE-derived OLs persist in the cortex [Ordaz et al 2019 (<https://doi.org/10.1038/s41467-019-11904-4>) and Li et al 2024 (<https://doi.org/10.1101/2023.12.01.569674>)]. Extending the current study to different rostro-caudal regions of the cortex would greatly improve the manuscript.

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Reviewer #3 (Public Review):

In the manuscript entitled "Embryonic Origins of Forebrain Oligodendrocytes Revisited by Combinatorial Genetic Fate Mapping," Cai et al. used an intersectional/subtractional strategy to genetically fate-map the oligodendrocyte populations (OLs) generated from medial ganglionic eminence (NKX2.1+), lateral ganglionic eminences, and dorsal progenitor cells (EMX1+). Specifically, they generated an OL-expressing reporter mouse line OpalinP2A-Flpo-T2A-tTA2 and bred with region-specific neural progenitor-expressing Cre lines EMX1-Cre for dOL and NKX2.1-Cre for MPOL. They used a subtractional strategy in the OpalinFlp::Emx1Cre::Nkx2.1Cre::RC::FLTG mouse line to predict the origins of OLs from lateral/caudal ganglionic eminences (LC). With their genetic tools, the authors concluded that neocortical OLs primarily consist of dOLs. Although the populations of OLs (dOLs or MP-OLs) from Emx1+ or Nkx2.1+ progenitors are largely consistent with previous findings, they observed that MP-OLs contribute minimally but persist into adulthood without elimination as in the previous report (PMID: 16388308).

Intriguingly, by using an indirect subtraction approach, they hypothesize that both Emx1-negative and Nkx2.1-negative cells represent the progenitors from lateral/caudal ganglionic eminences (LC), and conclude that neocortical OLs are not derived from the LC region. This is in contrast to the previous observation for the contribution of LC-expressing progenitors (marked by Gsx2-Cre) to neocortical OLs (PMID: 16388308). The authors claim that Gsh2 is not exclusive to progenitor cells in the LC region (PMID: 32234482). However, Gsh2 exhibits high enrichment in the LC during early embryonic development. The presence of a small population of Gsh2-positive cells in the late embryonic cortex could originate/migrate from Gsh2-positive cells in the LC at earlier stages (PMID: 32234482). Consequently, the possibility that cortical OLs derived from Gsh2+ progenitors in LC could not be conclusively ruled out. Notably, a population of OLs migrating from the ventral to the dorsal cortical region was detected after eliminating dorsal progenitor-derived OLs (PMID: 16436615).

The indirect subtraction data for LC progenitors drawn from the OpalinFlp-tdTOM reporter in Emx1-negative and Nkx2.1-negative cells in the OpalinFlp::Emx1Cre::Nkx2.1Cre::RC::FLTG mouse line present some caveats that could influence their conclusion. The extent of activity from the two Cre lines in the OpalinFlp::Emx1Cre::Nkx2.1Cre::RC::FLTG mice remains uncertain. The OpalinFlp-tdTOM expression could occur in the presence of either Emx1Cre or Nkx2.1Cre, raising questions about the contribution of the individual Cre lines. To clarify, the authors should compare the tdTOM expression from each individual Cre line, OpalinFlp::Emx1Cre::RC::FLTG or OpalinFlp::Nkx2.1Cre::RC::FLTG, with the combined OpalinFlp::Emx1Cre::Nkx2.1Cre::RC::FLTG mouse line. This comparison is crucial as the results from the combined Cre lines could appear similar to only one Cre line active.

Overall, the authors provided intriguing findings regarding the origin and fate of oligodendrocytes from different progenitor cells in embryonic brain regions. However, further analysis is necessary to substantiate their conclusion about the fate of LC-derived OLs convincingly.

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